PAPER_385 — Canonical 7-System UQFF Parameter Registry

This paper establishes that registry — the authoritative configuration table for UQFF validation

Session: *104 (Complete Re-Analysis — full 18-field per-system registry not formalized in prior papers)*

PAPER_385 — Canonical 7-System UQFF Parameter Registry

Source: grok*share*11254865.txt, lines ~6700–6850 (confirmed 9400–10322 in main()) **Section:** `MUGESystem` struct initializations for all 7 canonical validation systems **Session:** 104 (Complete Re-Analysis — full 18-field per-system registry not formalized in prior papers) **CP4 Class:** `Canonical7SystemUQFFParameterRegistryCalculator` (CP4 #36)

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1. Overview

PAPER*379 compared the compressed vs resonance MUGE outputs for all 7 canonical systems. PAPER*377 noted the 18-field CSV format. But no paper explicitly documented all **18 parameters for each of the 7 systems** as a canonical reference registry.

This paper establishes that registry — the authoritative configuration table for UQFF validation that all C++ and Python implementations must match.

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2. The 18-Field MUGESystem Parameter Set

The `MUGESystem` struct defines 18 fields for each astrophysical system:

Field	Symbol	Description	Units
1. name	—	System identifier	string
2. I	I	Electric current	A
3. A	A	Cross-sectional area	m^2
4. omega1	ω■	Upper frequency	rad/s
5. omega2	ω■	Lower frequency	rad/s
6. Vsys	V_sys	System volume	m^3
7. vexp	v_exp	Expansion velocity	m/s
8. t	t	System age	s
9. z	z	Redshift	—
10. ffluid	f_fluid	Fluid oscillation frequency	Hz
11. M	M	Total mass	kg
12. r	r	Characteristic radius	m
13. B	B	Magnetic field	T
14. Bcrit	B_crit	Critical magnetic field	T
15. rho_fluid	ρ_f	Fluid density	kg/m^3
16. g_local	g_local	Local gravitational surface acc.	m/s^2

Field	Symbol	Description	Units
17. M_DM	M_DM	Dark matter mass	kg
18. delta*rho*rho	δρ/ρ	Fractional density contrast	—

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3. Canonical Parameter Registry — All 7 Systems

System 1: SGR1745 — Magnetar SGR 1745-2900

Field	Value
I	1×10^{21} A
A	$3.142\times10^{\blacksquare}$ m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	4.189×10^{12} m^3
v_exp	1×10^{3} m/s
t	3.799×10^{10} s
z	0.0009
f_fluid	$1.269\times10^{-1\blacksquare}$ Hz
M	2.984×10^{30} kg
r	$1\times10^{\blacksquare}$ m
B	1×10^{10} T
B_crit	1×10^{11} T
ρ_f	$1\times10^{1\blacksquare}$ kg/m^3
g_local	1.991×10^{12} m/s^2
M_DM	$1\times10^{2\blacksquare}$ kg
δρ/ρ	0.1

System 2: Sag A — Sagittarius A (SMBH)

Field	Value
I	1×10^{23} A
A	2.813×10^{30} m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	$3.552\times10^{\blacksquare\blacksquare}$ m^3
v_exp	$5\times10^{\blacksquare}$ m/s
t	$3.786\times10^{1\blacksquare}$ s
z	0.0009
f_fluid	$3.465\times10^{-\blacksquare}$ Hz
M	$8.155\times10^{3\blacksquare}$ kg
r	1×10^{12} m
B	$1\times10^{-\blacksquare}$ T

Field	Value
B_crit	$1\times10^{-}$■ T
ρ_f	1×10^{-1}■ kg/m^3
g_local	5.443×10^{2} m/s^2
M_DM	1×10^{3}■ kg
δρ/ρ	0.01

System 3: Tapestry — Tapestry of Blazing Starbirth

Field	Value
I	1×10^{22} A
A	1×10^{3}■ m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	1×10■^{3} m^3
v_exp	1×10■ m/s
t	3.156×10^{13} s
z	0.0
f_fluid	1×10^{-12} Hz
M	1.989×10^{3}■ kg
r	3.086×10^{1}■ m
B	$1\times10^{-}$■ T
B_crit	$1\times10^{-}$■ T
ρ_f	1×10^{-21} kg/m^3
g_local	1.39×10^{-1}■ m/s^2
M_DM	1×10^{3}■ kg
δρ/ρ	0.01

System 4: Westerlund 2 — Westerlund 2 Star Cluster

Field	Value
I	1×10^{22} A
A	1×10^{3}■ m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	1×10■^{3} m^3
v_exp	1×10■ m/s
t	3.156×10^{13} s
z	0.0
f_fluid	1×10^{-12} Hz
M	1.989×10^{3}■ kg

Field	Value
r	3.086×10^{1}■ m
B	$1\times10^{-}$■ T
B_crit	$1\times10^{-}$■ T
ρ_f	1×10^{-21} kg/m^3
g_local	1.39×10^{-1}■ m/s^2
M_DM	1×10^{3}■ kg
δρ/ρ	0.01

Note: Tapestry and Westerlund 2 share identical parameters — they represent two systems in the same star-forming complex with equivalent physical scale.

System 5: Pillars of Creation

Field	Value
I	1×10^{21} A
A	2.813×10^{32} m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	3.552×10■■ m^3
v_exp	2×10^{3} m/s
t	3.156×10^{13} s
z	0.0
f_fluid	8.457×10^{-1}■ Hz
M	1.989×10^{32} kg
r	9.46×10^{1}■ m
B	1×10^{-10} T
B_crit	$1\times10^{-}$■ T
ρ_f	1×10^{-23} kg/m^3
g_local	2.979×10^{-10} m/s^2
M_DM	1×10^{32} kg
δρ/ρ	0.05

System 6: Rings of Relativity — Rings of Relativity Gravitational Lens

Field	Value
I	1×10^{22} A
A	1×10^{3}■ m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	1×10■■ m^3
v_exp	1×10■ m/s

Field	Value
t	$3.156\times10^{1■}$ s
z	0.01
f_fluid	$1\times10^{-■}$ Hz
M	$1.989\times10^{3■}$ kg
r	$3.086\times10^{1■}$ m
B	1×10^{-10} T
B_crit	$1\times10^{-■}$ T
ρ_f	$1\times10^{-2■}$ kg/m^3
g_local	$1.391\times10^{-1■}$ m/s^2
M_DM	$1\times10^{3■}$ kg
δρ/ρ	0.02

System 7: Student's Guide — Student's Guide to the Universe (Cosmological)

Field	Value
I	$1\times10^{2■}$ A
A	$1\times10■^{2}$ m^2
ω■	1×10^{12} rad/s
ω■	9.99×10^{11} rad/s
V_sys	$1\times10■^{0}$ m^3
v_exp	$3\times10■$ m/s
t	$4.35\times10^{1■}$ s
z	0.0
f_fluid	$1\times10^{-1■}$ Hz
M	$1\times10■^{3}$ kg
r	$1\times10^{2■}$ m
B	$1\times10^{-1■}$ T
B_crit	$1\times10^{-1■}$ T
ρ_f	$1\times10^{-2■}$ kg/m^3
g_local	6.67×10^{-33} m/s^2
M_DM	$1\times10■^{3}$ kg
δρ/ρ	0.001

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4. CSV Format (18-Field Standard)

The canonical CSV header for UQFF system configuration files:

```
name,I,A,omega1,omega2,Vsys,vexp,t,z,ffluid,M,r,B,Bcrit,rho_fluid,g_local,M_DM,delta_rho_rho
```

```
sgr1745,1e21,3.142e8,1e12,9.99e11,4.189e12,1e3,3.799e10,0.0009,1.269e-14,2.984e30,1e4,1e10,1e11,1e15,1.991e12,1e28,0.1
```

```
sagA,1e23,2.813e30,1e12,9.99e11,3.552e45,5e6,3.786e14,0.0009,3.465e-8,8.155e36,1e12,1e-5,1e-4,1e-19,5.443e2,1e38,0.01
```

```
tapestry,1e22,1e35,1e12,9.99e11,1e53,1e4,3.156e13,0.0,1e-12,1.989e35,3.086e17,1e-9,1e-8,1
e-21,1.39e-15,1e36,0.01
westerlund,1e22,1e35,1e12,9.99e11,1e53,1e4,3.156e13,0.0,1e-12,1.989e35,3.086e17,1e-9,1e-8
,1e-21,1.39e-15,1e36,0.01
pillars,1e21,2.813e32,1e12,9.99e11,3.552e48,2e3,3.156e13,0.0,8.457e-14,1.989e32,9.46e15,1
e-10,1e-9,1e-23,2.979e-10,1e32,0.05
rings,1e22,1e35,1e12,9.99e11,1e54,1e5,3.156e14,0.01,1e-9,1.989e36,3.086e17,1e-10,1e-9,1e-
28,1.391e-14,1e38,0.02
student_guide,1e24,1e52,1e12,9.99e11,1e80,3e8,4.35e17,0.0,1e-18,1e53,1e26,1e-15,1e-14,1e-
26,6.67e-33,1e53,0.001
```

5. System Scale Spectrum

System	Class	r (m)	M (kg)	Physics Domain
SGR1745	Magnetar	1×10■	2.984×10^{30}	NS surface gravity
Sag A*	SMBH	1×10^{12}	8.155×10^{3}■	Galactic center
Pillars	GMC	9.46×10^{1}■	1.989×10^{32}	Stellar nursery
Tapestry	LMC-scale	3.086×10^{1}■	1.989×10^{3}■	Star-forming complex
Westerlund 2	Cluster	3.086×10^{1}■	1.989×10^{3}■	OB star cluster
Rings	Lens	3.086×10^{1}■	1.989×10^{3}■	Gravitational lens
Student's Guide	Cosmological	1×10^{2}■	1×10■^{3}	Observable universe

The 7 systems span **22 orders of magnitude** in radius (10■ m to 10^{2}■ m) and **23 orders in mass** (10^{30} kg to 10■^{3} kg), making this the most comprehensive UQFF multi-scale validation suite in the codebase.

6. Canonical Computed Results Summary

For reference — results from PAPER*379 (full comparison) and PAPER*382/384 (per-term):

System	Compressed MUGE (m/s^2)	Resonance MUGE (m/s^2)	Ratio
SGR1745	1.782×10^{3}■	$1.773\times10^{-}$■	10■■
Sag A*	2.966×10^{3}■	4.105×10^{2}■	~10■
Tapestry	$\sim GM/r^2$	fluid-dominated	converge
Westerlund 2	$\sim GM/r^2$	fluid-dominated	converge
Pillars	$\sim GM/r^2$	fluid-dominated	converge
Rings	$\sim GM/r^2$	fluid-dominated	converge
Student's Guide	$\sim GM/r^2$	fluid-dominated	converge

7. References Within Codebase

- PAPER_371: Resonance MUGE framework — 12-term formula
- PAPER_372: Compressed MUGE framework — 8-term formula
- PAPER_379: Dual-model 7-system comparison table
- PAPER_377: `load_muge_systems()` CSV parser (18-field format)
- `WormholeMUGETermImplSafetyCalculator` (CP4 #26): 7-system dict

`MUGESuperconductive12TermResonanceCalculator` (CP4 #14): uses sagA_dataset

Source: grokshare11254865.txt lines ~6700–6850 + lines ~9400–10322 (main() C++ impl.) | Session 104 | First formal 18-field canonical parameter registry for all 7 UQFF validation systems

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